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Reactive and proactive agency in GenAI-assisted speaking practice: task fulfilment, boundary testing, and ownership claiming among young adolescent EFL learners

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ABSTRACT

Purpose: Speech-enabled generative artificial intelligence (GenAI) chatbots offer opportunities for second-language speaking, yet participation does not necessarily give learners conversational control. This study examines how young adolescent EFL learners enacted reactive and proactive agency during repeated GenAI-mediated speaking interactions and how personal, contextual, and relational resources constrained that agency.

Design: A qualitative study with repeated speaking encounters over 12 weeks involved 11 Grade 8 EFL learners at a Chinese secondary school. Across four sessions, 44 learner-chatbot interactions generated 746 dialogue rounds. Abductive thematic analysis moved iteratively between time-ordered screen records, stimulated-recall interviews, teacher-observation notes, Bandura's four agency features, and a framework of personal, contextual, and relational resources.

Findings: Reactive agency regulated action towards teacher-assigned task fulfilment through preparation, help-seeking, clarification, repair, and topic replacement. Proactive agency arose when learner-generated purposes organised boundary testing and ownership claiming, including resistance, reciprocal questioning, and topic redirection. Evidence for forethought and self-reflectiveness varied across dialogue rounds. These configurations describe how learners oriented their purposes and actions, without implying differences in agency levels or learning quality. Linguistic and processing demands, task and technological conditions, available support, and the chatbot's recurrent control of questions and topics shaped whether learners could sustain participation or influence what happened next.

Originality: The study contributes a learner-control account of agency that distinguishes speaking opportunity and fluent participation from influence over conversational purpose and direction. By connecting purpose, regulation, and conversational consequence, it explains how young learners manage assigned activity and make their purposes consequential within teacher-structured GenAI speaking practice.

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1. Introduction

Speech-enabled generative artificial intelligence (GenAI) chatbots have become a promising resource for EFL speaking because they can give learners repeated opportunities to interact in English. Studies with elementary and adolescent EFL learners have reported improved speaking performance after chatbot-supported activities (Tai and Chen 2024, 2025). Other research has associated GenAI-speaking practice with oral-proficiency development, willingness to communicate, enjoyment, reduced anxiety, and engagement (Guan, Zhang, and Gu 2025; Wang, Li, and Zou 2025; Wu, Hapsari, and Huang 2026).

A learner may speak fluently and complete a chatbot activity while remaining a respondent to chatbot-initiated questions. This participation-control paradox is not resolved by positive learning and affective outcomes alone. Speaking scores show whether performance changed, while measures of willingness to communicate, enjoyment, anxiety, and engagement show how learners appraised an activity. Neither type of evidence can establish why a learner chose to continue a conversation, ask for clarification, repeat an answer, accept the chatbot's direction, redirect the topic, or end an exchange. These actions may reflect different purposes and different attempts to maintain or control the conversation. Research has also identified motivational factors associated with GenAI-speaking practice and interactional strategies used when chatbots produce problematic responses (Choi 2026; Choi and Oh 2026; Guan, Zhang, and Gu 2025; Wang, Li, and Zou 2025). What remains unclear is how learners' motivations, purposes, and strategies operate together as they manage a live GenAI-mediated speaking interaction.

This gap is especially consequential for young adolescent EFL learners, whose opportunities to exercise control may be shaped by their linguistic resources, prior experience, and the chatbot's responses. Existing school-based research has primarily evaluated learning and affective outcomes following GenAI-speaking activities (Tai and Chen 2024, 2025). Early process-oriented studies have begun to examine repair and turn-taking in L2 learners' interactions with ChatGPT (Choi 2026; Choi and Oh 2026). Yet little is known about how young learners sustain, negotiate, redirect, or withdraw from repeated GenAI-mediated speaking interactions, and how they explain the purposes underlying these actions.

Learner agency offers greater explanatory value for the present question because it connects learners' purposes with their capacity to influence the unfolding interaction. Its focus on purposeful influence complements engagement (involvement), willingness to communicate (readiness to speak), self-regulation (management of goal-directed action), and autonomy (responsibility for learning). These overlapping constructs do not make participation alone evidence of conversational control. Reeve and Tseng (2011) found that agentic engagement explained achievement variance beyond behavioural, emotional, and cognitive engagement. In language education, Papi and Hiver (2025) likewise emphasise learners' deliberate efforts to shape learning opportunities. These accounts support examining how learners make their purposes consequential within an interaction. Bandura's (2001, 2006) account directs attention to *intentionality*, *forethought*, *self-reactiveness*, and *self-reflectiveness*: the purposes learners pursue, the actions they anticipate, the ways they regulate action, and the evaluations they make during and after an interaction. This framework treats learner control as a process that must be evidenced through the relationship between purpose, action, chatbot response,

and subsequent learner action. Lai et al. (2026) likewise conceptualised agency with GenAI as learners' negotiation of sociotechnical affordances and constraints. Their work focused on Chinese university learners' self-directed informal language learning, leaving open how agency operates among younger learners in teacher-assigned GenAI-speaking tasks. Following the distinction between externally oriented and personally directed action in language-education research (Ruohotie-Lyhty and Moate 2015), we use reactive agency for purposeful regulation of an assigned activity and proactive agency for introducing or redirecting a local conversational purpose.

To address this process gap, this study investigates how young adolescent EFL learners enact agency while sustaining and shaping GenAI-mediated speaking interactions. It addresses the following research questions:

RQ1. How did young adolescent EFL learners enact reactive and proactive agency in GenAI-mediated speaking interactions?

RQ2. How did personal, contextual, and relational resources enable or constrain young adolescent EFL learners' agency in GenAI-mediated speaking tasks?

This study advances understanding of pedagogical innovation in GenAI-mediated speaking by treating learner control as a criterion for evaluating these activities. Drawing on linked interaction records, stimulated-recall interviews, and teacher observation notes, it examines how reactive regulation sustains assigned activities, how proactive intentionality introduces conversation purposes, and how personal, contextual, and relational resources shape the enactment of both types of agency. This process-based account distinguishes opportunities to speak and complete assigned tasks from learners' capacity to influence conversational direction. It proposes a dialogue-round analysis based on how learner purposes relate to the assigned activity and the chatbot-led sequence. The study contributes an account of how young learners exercise control during spoken interactions with a GenAI chatbot.

2. Literature review and conceptual framework

2.1. What GenAI-speaking outcome evidence establishes

GenAI-speaking research has primarily examined three forms of outcome: speaking performance, learners' affective and acceptance-related responses, and engagement under different forms of interaction and feedback. Across these strands, studies report promising learning and experience outcomes, although the evidence concerns the results of particular GenAI-speaking activities rather than the processes through which learners sustain, regulate, or reshape a conversation (Guan, Zhang, and Gu 2025; Tai and Chen 2024, 2025; Wang, Li, and Zou 2025; Wu, Hapsari, and Huang 2026; Zou et al. 2026).

Empirical findings illustrate both the promise and the limits of this outcome-focused evidence. Guan, Zhang, and Gu (2025) found a modest post-test speaking advantage for individual GenAI practice over teacher-led group practice, although the intervention also altered learners' speaking opportunities, feedback, and attention. Tai and Chen (2024, 2025) similarly reported immediate speaking advantages for elementary and adolescent learners using GenAI-supported interaction modes; their activities combined chatbot use with structured topics, worksheets, and individual or paired participation. Research on

affect and acceptance extends this evidence by showing that learners' AI-specific willingness to communicate is associated with perceived usefulness and ease of use, and that short-term outcomes vary across proficiency groups (Wang, Li, and Zou 2025). Anxiety, enjoyment, and speaking experience also vary across AI-mediated activities and conversational topics (Wu, Hapsari, and Huang 2026; Yetkin and Aslan 2026). Studies of feedback and engagement report further differences: Zou et al. (2026) found stronger grammatical-accuracy and in-class willingness-to-communicate gains in a teacher-informed, lower-interactivity feedback condition than in more open dialogic modes, while Min, Gao, and Zhang (2026) found higher behavioural, cognitive, and social engagement under teacher guidance, with cognitive engagement also differing by task mode. Evidence from younger learners is particularly relevant. Tai and Chen's (2024, 2025) studies show that elementary and adolescent learners can benefit from GenAI-supported speaking, yet their outcome measures and interviews do not show how learners' purposes, strategies, and control of the dialogue develop across repeated encounters. Wang and Derakhshan (2026) examined engagement's psychological correlates through network analysis, while Wang, Gao, and Liu (2026) traced enjoyment and engagement across three survey waves in AI-mediated informal English learning. These studies extend evidence about participation without directly establishing whose purpose organises a spoken exchange. Taken together, the literature establishes whether learners improved, how they appraised an activity, and how outcomes differed across conditions. It provides less evidence about how learners respond when a chatbot's turn, topic, or feedback does not serve their immediate communicative purpose.

2.2. From interactional difficulty to learner control

Emerging process-oriented research has begun to illuminate how learners manage difficulties during GenAI-mediated speaking. Choi (2026) showed how L2 learners used third-position repair when chatbot responses revealed misunderstandings, while Choi and Oh's (2026) longitudinal conversation analysis traced changes in learners' turn-taking practices across repeated ChatGPT role-play sessions. These studies demonstrate that learners actively repeat, clarify, reformulate, and adjust their contributions to sustain an interaction. However, repair and turn-taking primarily reveal what learners do within observable conversational sequences; they do not necessarily establish the purposes underlying these actions or whether continuing the exchange serves the learner's own communicative goals. This limitation is also apparent in broader GenAI-language-learning research. Guan, Zhang, and Gu (2025) found that perceived usefulness and access to individual conversational practice did not necessarily ensure sustained voluntary engagement, suggesting that speaking opportunities acquire value through learners' appraisals and regulation of their participation. Similarly, using think-aloud data, screen recordings, and chat histories, Kim et al. (2026) identified qualitatively different patterns of student-GenAI interaction associated with learners' existing AI literacy and writing performance. Although their study concerned university writing rather than young learners' speaking, it demonstrates how outcome measures and aggregate indicators of participation can conceal different ways of organising activity with GenAI. These studies establish the value of process evidence while leaving unresolved how learners' purposes, anticipations, actions, and evaluations combine as they negotiate control of an unfolding spoken interaction.

Participation and task completion should therefore not be treated automatically as evidence of learner agency. A learner may answer questions, repeat an utterance, or request clarification to meet a teacher-assigned requirement while the chatbot continues to control the topic and turn sequence. Conversely, refusal, topic redirection, or an attempt to close an exchange may appear to signal disengagement but can also constitute purposeful efforts to protect, manage, or reorient the interaction. Determining the agentic significance of such actions requires a framework connecting learners' purposes and anticipated outcomes with their regulation and evaluation of action. The next section therefore develops a learner-agency framework based on Bandura's (2001, 2006) four features of human agency and a situated resource perspective.

2.3. Learner agency as an interactional and resource lens

Current empirical studies portray learner agency as situated and uneven, with learners drawing on technological and educational resources in different ways. In out-of-class language learning, Lai and Gu (2011) found that learners selected and combined technologies according to their goals, beliefs, and perceived needs, resulting in different patterns of self-regulation. Soyoof et al.'s (2023) scoping review identified agency and digital literacies as an emerging focus in informal digital language learning, particularly in learners' critical evaluation and appropriation of digital resources. Evidence from secondary education further supports this situated account. Based on survey data from 1,470 students in the Netherlands and Belgium, Schoots-Snijder, Tigelaar, and Admiraal (2026) identified 11 agency profiles associated with different patterns of academic performance, well-being, and stress. The variation within and across these profiles shows that students mobilise their will, skills, and environmental resources in different combinations. Agency therefore varies across learners and learning conditions, even among students participating in the same educational setting. Ruohotie-Lyhty and Moate's (2015) language-learning biographies distinguish reactive orientations towards others' expectations from proactive pursuit of personally meaningful goals. Papi and Hiver's (2025) proactive language learning theory foregrounds strategic seeking of input, interaction, feedback, and information; Lin and Huang (2026) identify bidirectional proactive and reactive pathways in online language classes. These studies support a situated distinction without implying fixed learner types or an inevitable developmental sequence.

Research involving AI chatbots further locates agency in learners' experimentation with technological affordances and their regulation of AI-supported activity. Using survey responses from 867 Chinese EFL learners and interviews with 20 participants, Liu, Darvin, and Ma (2025) found that hands-on experimentation shaped perceptions of chatbot usefulness, while negotiation of technological affordances and constraints influenced learners' intention to continue using the systems. Learners appropriated chatbots as tutors and conversational partners and refined their prompting strategies in response to limitations in chatbot output. Related classroom research with 36 Korean primary learners found that pedagogical, technological, and social chatbot affordances functioned as opportunities or constraints according to each learner's perceptions and experiences (Jeon 2024). Lai et al. (2026) provide an empirical examination of agency in GenAI-supported language learning. Interviews and interaction records from 17 Chinese undergraduate English majors, followed by a validation survey of 646 learners,

produced five dimensions of agency: performance proxy, learning proxy, spatial, collective, and shared agency. The findings distinguished direct task delegation from deliberate engagement with GenAI output for learning and showed how learners extended learning across settings, expanded possible practices, and coordinated AI output with human and technological resources. Longer and more frequent GenAI use was positively associated with all five self-reported dimensions, while higher English proficiency was associated with less collective agency and more shared agency. These findings show that agency differs in purpose, form, and resource demands. Current evidence centres on university learners' informal GenAI use, self-reported agency dimensions, and young learners' perceptions of non-generative chatbots. For the present analysis, task fulfilment, boundary testing, and ownership claiming are abductively constructed configurations concerning task maintenance, testing the chatbot's limits, and attempts to influence conversational direction. Task completion alone is insufficient: agency requires evidence of purposeful choice or adjustment. Bandura's self-reactiveness concerns regulation in both reactive and proactive dialogue rounds and is distinct from the reactive agency label. Informed by Lai et al. (2026) but distinct from their five-dimensional typology, this analytic synthesis groups resources as personal (learner capacities), contextual (task and technological conditions), and relational (distribution of conversational initiative).

3. Method

3.1. Research context and participants

The sample comprised 11 Grade 8 EFL learners aged 13-14 from one junior secondary school in Suzhou, China via convenience sampling. All participants had begun learning English in Grade 3 and were in approximately their sixth school year of English study. Grade 8 was selected because learners had completed their first year of junior secondary English and were not yet preparing for the senior-high entrance examination. The Cambridge A2 Key for Schools test was used as the pre-test; the group mean was 109.27 (SD = 13.42; range = 87–128). Participants were invited if they were willing to speak with a mobile application, were recommended by their English teachers based on classroom performance, had parental consent after being informed about the study, and were willing to bring a personal mobile phone with the application installed. Teacher recommendation, willingness to speak, and personal-phone access may have favoured learners with stronger classroom performance, confidence, or technological access. Nine participants reported no previous chatbot experience and two reported some experience. The Institutional Review Board at City University of Macau approved the study under reference 2026-RE-16. Names and account identifiers were replaced with participant codes P01-P11, and potentially identifying contextual details were generalised. Raw voices, recordings, and identifiable app records are restricted to the research team and will not be placed in a public repository.

3.2. The GenAI-powered mobile application

Hi Echo is a speech-enabled AI conversation application developed by NetEase and presented in partnership with the British Council. It accepted learners' spoken English, displayed recognised text, and generated spoken and written replies. Learners could select an avatar

and voice, view or hide the on-screen transcript, and receive written feedback after the conversation (Figure 1). The partnership provided institutional context for the platform's selection; the study did not treat it as evidence of pedagogical effectiveness or learner agency. During data collection, the study used Hi Echo version 3.7.6, powered by NetEase Youdao's ZiYue model. The app was configured for the middle-school stage, beginner difficulty, and standard speech speed; learners selected the available accent and avatar.

The pedagogical routine combined topic preparation with open conversation. Classroom instruction introduced each topic, learners could prepare questions, and each learner was asked to sustain an individual conversation for more than ten minutes within the 15-minute maximum. Observation records documented behavioural and affective responses and any help-seeking that occurred. This arrangement placed learner-generated goals within a teacher-defined topic and duration.

3.3. Data collection procedures

Data were collected across four sessions (Sessions 1-4) over 12 weeks in the second semester of the 2025–2026 academic year. Each session included approximately one hour of topic-related instruction delivered by the postgraduate research assistant (RA) and an individual chatbot conversation of up to 15 min, yielding 44 scheduled learner-chatbot interactions. An interaction comprised one learner's complete scheduled chatbot conversation without using the chatbot translation function. Learners entered prompts orally while being allowed to consult prepared written questions. They could consult the on-



Figure 1. Example of the app's interactional configuration.

screen transcript and receive teacher or peer assistance. Topics were aligned with textbook reading passages and allocated across the four sessions as follows: Session 1 covered Chinese places, culture, travel, and food; Session 2 covered weekend leisure, sports and fitness, and food; Session 3 covered regional culture, films, and friendship; and Session 4 covered food and weekend leisure.

Time-ordered learner-chatbot interactions, stimulated-recall interviews, and teacher-observation notes formed the three main data sources. During the interactions on the mobile phone, the first author and the RA with two years of teaching experience sat near the learners and noted down contemporaneous observations of preparation, hesitation, laughter, help-seeking, recognition difficulties, responses to lengthy chatbot turns, and topic redirection. These notes were kept as the teacher-observation notes. The 44 learner-chatbot interactions were screen recorded and transcribed, resulting in a corpus of 746 dialogue rounds consisting one initiation and one response between the learner and chatbot (mean 16.95 rounds per interaction). All interactions were retained in the corpus. Screen recordings were the authoritative source for displayed wording and round boundaries; Optical Character Recognition (OCR) and audio records were checked manually against them. Live Automatic Speech Recognition (ASR) errors were retained because they formed part of the interaction. The two observers used the shared focal categories listed above, and their notes served as contextual evidence rather than independently scored ratings.

After all 44 AI-chat interactions had been completed, one stimulated-recall interview was conducted with each of the 11 learners while they watched recordings from their interactions. Each interview lasted 5–20 min. The RA conducted the interviews in person, while the first author joined online and asked follow-up questions when necessary. The interview prompts largely came from two sources: the points raised in teacher-observation notes and the transcripts of the participants' spoken output during their interactions with the GenAI-powered chatbot. Example questions about *intentionality* include: 'Did you have a "secret mission" or a goal for this chat/question/answer? What was the goal to say this sentence?' Example questions for *forethought* include: 'What did you do when you saw that? Did you copy it, ask for a simpler version, or ignore it?'. Example questions for *self-reactiveness* include: 'When the chatbot gave you an answer, how did it make you feel? Did that feeling change what you did next?' Finally, example questions for *self-reflectiveness* include: 'Looking back at your whole conversation, do you think you did a good job? What makes you say that?' Interviews were conducted in Mandarin, the learners' preferred language. Interview recordings were transcribed and analysed in Chinese; English translations were prepared only for excerpts reported in this article so that meaning was retained during analysis.

3.4. Data analysis

The study used thematic analysis with an abductive logic to examine agency across the three linked qualitative datasets: the transcription of 44 scheduled learner-chatbot interactions (LCI), stimulated-recall interviews (SRI), and teacher-observation notes (TON). Analysis moved iteratively between datasets (see Table 1), Bandura's (2001, 2006) theory of human agency, and the resource grouping described in Section 2.3. Following Braun and Clarke's (2006) six recursive phases, two authors familiarised themselves with

and indexed linked records; independently generated codes; collated codes into agency configurations and resource constraints; reviewed them against coded extracts, full records, and within-case variation; defined and named final configurations; and selected extracts for reporting. The first author's involvement in observation and interviewing provided contextual familiarity while also creating potential influence on the analysis; comparison between the two authors and the retained links to source records supported reflexive scrutiny. The authors compared coding decisions and resolved differences through discussion; this supported agreed coding decisions. Ambiguous or contradictory cases were retained, and intentionality claims were qualified when recall evidence was unavailable.

The coding unit across all three datasets was a phrase or sentence. In the interaction transcripts, each coded segment was interpreted within its dialogue round and the surrounding interaction. Coded segments from the stimulated-recall interviews and teacher-observation notes were linked by participant and session and, where possible, to a dialogue round. Codes were compared across the three sources and interpreted in relation to the theoretical framework to develop the final themes abductively. Bandura's features and the resource lenses informed coding, whereas the three agency configurations were generated through comparison across the coded data. For RQ1, codes identified three agency configurations in learner-chatbot interactions: task fulfilment, boundary testing, and ownership claiming. *Forethought*, *self-reactiveness*, and *self-reflectiveness* were examined across the three configurations, whereas *intentionality* was coded only between reactive and proactive intentions. For RQ2, constraints were coded within the framework of limitations in personal, contextual, and relational resources. Table 1 and Appendix A document the coding framework and its links to the three sources.

4. Findings

Drawing on Bandura's (2001, 2006) framework, the analysis distinguished reactive and proactive agency according to the origin of learners' overarching *intentionality*. Reactive agency involved regulating action towards the teacher-assigned speaking task. Proactive agency occurred when learners introduced a self-generated *intentionality* through

Table 1. Illustrative coding and triangulation audit.

Code	Coding threshold	Linked source	Illustrative audit link
Intentionality	A stated purpose for the episode or selected action	SRI + LCI	P11 described being able to generate topics and questions rather than follow a fixed agenda.
Forethought	Stated advance preparation or anticipated next action/outcome	SRI + LCI	P03 confirmed preparing questions before the conversation.
Self-reactiveness	Observable enactment, monitoring, repair, or adjustment	LCI + TON	P02 replaced an unresponsive food prompt with 'What about dumpling?'
Self-reflectiveness	Evaluation of strategy, system response, outcome, or available resources	SRI + LCI	P03 evaluated vocabulary and grammar limits after a recognition difficulty.
Resource constraint	Personal, contextual, or relational condition limiting the next action	SRI + LCI + TON	Long AI turns and lexical difficulty constrained P08's available responses.

Note. This table provides illustrative examples of coding and triangulation; it is not an exhaustive codebook.

boundary testing or ownership claiming. The classification therefore concerned who determined the purpose and direction of the interaction. *Forethought*, *self-reactiveness*, and *self-reflectiveness* operated across all three configurations when supported by the linked datasets. The following presentation of the data will follow the coding scheme: learner-chatbot interactions (LCI), stimulated-recall interviews (SRI), and teacher-observation notes (TON), participant (P01–P11), session (S1–S4), and, where applicable, the preserved interaction record (U###).

4.1. Reactive agency as task fulfilment

Reactive agency took the form of task fulfilment when learners regulated their participation within the assigned topic and duration. Learners exercised control over how they prepared for the task, managed linguistic and technological difficulties, and evaluated their performance.

Forethought supported task fulfilment through advance planning, cognitive preparation, rehearsal, and anticipated compensatory strategies. Written questions were documented across multiple sessions for P01, P03, P06, and P08 to prepare for the upcoming tasks (SRI-P03-S1; TON-P01-S1 to TON-P01-S4; TON-P03-S1 to TON-P03-S4; TON-P06-S1 to TON-P06-S4; TON-P08-S1 to TON-P08-S4). Preparation also involved allocating additional time to formulate responses. For example, P10 'needed a relatively long time to think and prepare his expression' before responding (TON-P10-S3). P01 quietly rehearsed her responses several times before answering to minimise recognition errors, a practice documented by the teacher in first three sessions (TON-P01-S1 to TON-P01-S3). Learners also anticipated linguistic problems and prepared fallback strategies. P02 described preparing topics before class while recognising that he might not be able to produce all the planned language. His solution was to 'change to something simpler' when he could not express an idea (SRI-P02-S2). This strategy became visible when he sought an English translation of Zhongshan Ling (The Sun Yat-sen Mausoleum), judged the translated expression too difficult, and inserted the Chinese place name into an English response so that the sightseeing topic could continue (TON-P02-S1; LCI-P02-S1/U013). P04 similarly reported replacing an unavailable word with 'a simpler word he could say' or using 'I don't know' (SRI-P04-S3). *Forethought* thus reflected learners' readiness to sustain participation, which extended beyond the preparation of written questions.

Self-reactiveness was the most extensively evidenced feature of reactive agency because learners repeatedly adjusted their actions in response to linguistic difficulty, recognition errors, and conversational drift. Error recovery included accommodating the chatbot's interpretation, repeating or reformulating input, slowing speech, and replacing unsuccessful questions. When 'climbing' was recognised as the name 'Kimi', P04 accepted the chatbot's version and treated Kimi as an imaginary friend so that the conversation could continue (TON-P04-S1; LCI-P04-S1/U018). P08 similarly followed the chatbot's misrecognised Beijing-history topic instead of interrupting the exchange to correct it (TON-P08-S1; LCI-P08-S1/U017). P01 described this accommodation strategy as continuing 'by answering its question' (SRI-P01-S3). When accommodation was insufficient, learners attempted more direct repairs. P10 repeated a question after a recognition failure, while P11 described his usual response as to 'say it again' (TON-P10-S1; SRI-P11-S4). P09 reduced her speaking rate after noticing recognition errors; by Session 4, slowing

down had become a routine repair that no longer interrupted the interaction (TON-P09-S2; TON-P09-S4; LCI-P09-S4/U038). Learners also steered the conversation back towards the assigned task. When 'robot' was displayed as 'noodle', P01 briefly acknowledged the new thread before stating, 'I like noodles, but I want to ask you something about robots' (TON-P01-S1). P11 similarly participated in a misrecognised exchange, stated that he was not interested in the new topic, and returned to his original question (TON-P11-S3; LCI-P11-S3/U033). Strategic adjustments to response scope provided a further means of maintaining participation. P08 answered only the part of a question that he understood and used Chinese or 'I don't know' for the remainder (TON-P08-S2; LCI-P08-S2/U018). P06 requested permission to use Chinese for an unavailable name, while P04 made his needs explicit through 'I don't know', 'Use simple words', and 'Please say it again' (TON-P06-S1; LCI-P06-S1/U016; LCI-P04-S2/U019). Across these interactions, *self-reactiveness* enabled learners to recover from breakdowns, regulate the scope of their responses, and continue participating without changing the assigned purpose.

Self-reflectiveness informed task fulfilment when learners evaluated errors, recognised their strategies, and compared their performance across sessions. Learners did not treat every recognition error as requiring correction. P10 continued speaking when an error did not affect the central meaning of his contribution, while P01 pursued repair only when the interaction could no longer recover her intended meaning (TON-P10-S4; LCI-P10-S4/U042; TON-P01-S4; LCI-P01-S4/U017). Interview accounts also made learners' awareness of their strategies explicit. P02 explained that he either changed an unavailable expression 'to something simpler' or 'followed what it said' after a recognition error (SRI-P02-S3). The clearest reflective evidence concerned perceived development across sessions. P04 stated, 'I can speak now; unlike before, I no longer stop when I forget' (SRI-P04-S4). His Session 4 interaction supported this evaluation through faster and more complete responses, the use of 'me too', and the absence of teacher assistance (TON-P04-S4; LCI-P04-S4/U040). *Self-reflectiveness* therefore helped learners determine whether errors mattered, whether compensatory strategies remained useful, and whether their capacity to sustain the assigned interaction had changed.

4.2. Proactive agency as boundary testing and ownership claiming

Proactive agency occurred when learners extended the interaction beyond maintaining the assigned activity and introduced a self-generated conversational purpose. Two configurations were identified. Boundary testing positioned the chatbot as an object of inquiry whose capabilities, limitations, and persona could be examined. Ownership claiming positioned the learner as a conversational initiator who selected topics, questions, and subsequent conversational directions.

4.2.1. Boundary testing

Boundary testing occurred when learners introduced unexpected content to investigate the chatbot's limits and responses. This boundary testing examined its linguistic flexibility, recognition capability, topic enforcement, and responses to provocative content.

Forethought in boundary testing involved asking a question expected to produce a particular type of chatbot response, although evidence of advance planning was uneven. P07 asked, 'Can you speak Chinese?' and later presented Chinese dish names followed by 'Can

you say them in English?' (TON-P07-S1; LCI-P07-S1/U012; TON-P07-S2; LCI-P07-S2/U022). P05 used a comparable dish-name translation question, while P11 asked, 'Do you know my friend's name?' to test the chatbot's social knowledge (TON-P05-S2; LCI-P05-S2/U023; TON-P11-S3; LCI-P11-S3/U033). These questions may request a particular response, but their wording alone does not establish advance planning; improvised experimentation remains a possible interpretation.

Self-reactiveness enabled learners to sustain boundary testing through persistence and reintroduction of boundary-testing content, accommodation of recognition failures, and regulation of turn distribution. When P05's first two attempts to say 'rob a bank' were misrecognised, he briefly answered the chatbot's resulting questions before reintroducing the phrase until it was recognised (TON-P05-S4; LCI-P05-S4/U044). P07 similarly returned to 'I don't know, I want to rob' after the conversation had moved away from his original statement (TON-P07-S3; LCI-P07-S3/U035). In both cases, learners temporarily accommodated the chatbot's responses before returning to the provocative content; without recall evidence linked to these dialogue rounds, a testing purpose remains an interpretation rather than an established intention. Learners also regulated who was responsible for producing the next contribution. P07 repeatedly used 'I don't know, what about you?' to redirect the chatbot's questions back to it (TON-P07-S1; LCI-P07-S1/U012). This strategy interrupted the chatbot's question-led sequence and prompted it to produce content that the learner could observe and evaluate. *Self-reactiveness* therefore sustained boundary testing by helping learners adjust to interactional disruptions without abandoning their boundary testing and by enabling them to redistribute conversational responsibility.

Self-reflectiveness in boundary testing centred on evaluations of the chatbot's responses, capabilities, and relational character. After the chatbot responded seriously to 'rob a bank' without offering moral evaluation, P05 discontinued the topic because he had no substantive follow-up prepared (TON-P05-S4; LCI-P05-S4/U044). He later described the chatbot as an 'old friend' and the interaction as 'interesting' (SRI-P05-S4). P07's evaluation changed across repeated interactions. Although he initially found the chatbot interesting, he later concluded that it was 'nothing particularly special' (SRI-P07-S2). These evaluations concern the overall experience and do not independently establish the purpose or reflective value of each instance of boundary testing.

4.2.2. Ownership claiming

Ownership claiming occurred when learners sought to determine what the interaction discussed and how it subsequently developed. They selected topics, posed learner-generated questions, rejected unwanted directions, and redistributed responsibility for the next turn.

Forethought supported ownership claiming by connecting prepared content with anticipated topic extension. P03 confirmed that she had prepared questions, but her Session 4 record distinguished these questions from additional requests for advice and information generated as the interaction developed (SRI-P03-S4; TON-P03-S4; LCI-P03-S4/U036). P11 also used written preparation before moving towards increasingly specific questions about the chatbot's activities (TON-P11-S4; LCI-P11-S4/U045). Planning therefore provided an initial conversational direction and possible avenues for extension without determining the complete interaction in advance.

Self-reactiveness provided the strongest evidence of ownership claiming because learners enacted control through refusals, topic replacement, reciprocal questioning, and

turn-by-turn redirection. P03 directly rejected an unwanted conversational direction: 'I don't want to talk about this. I want to talk about something about you' (TON-P03-S3; LCI-P03-S3/U026). The displayed chatbot reply was 'Sure, let's talk about movies I love. How about you? What's your favourite movie genre?', followed by P03's 'What's you movie do you like best?' The chatbot then named 'Inception' before asking another question (U026). This sequence shows local uptake of her alternative, although the observation note also links her redirection to difficulty understanding the preceding question. In another interaction, P03 responded to the chatbot's request for an explanation with 'Can you guess why I … ?', thereby transferring interpretive responsibility to the chatbot (TON-P03-S4; LCI-P03-S4/U036). She also used requests for advice and information to determine what the chatbot should contribute next. These actions repositioned the learner from respondent to questioner and influenced the subsequent conversational direction. The agentic meaning of redirection depended on the purpose it served. Returning to the assigned topic after a recognition failure supported reactive agency by sustaining task fulfilment. By contrast, rejecting the chatbot's direction and introducing a topic-relevant, learner-selected alternative asserted conversational ownership and enacted proactive agency. Topic change alone was therefore insufficient evidence of proactive agency.

Self-reflectiveness in ownership claiming was supported by a limited set of evaluations of adaptation and growing conversational control. For example, Participant 11 (P11) reported that he could 'gradually adapt' and described a shift from being 'nervous at first' to feeling 'somewhat excited' (SRI-P11-S3). His later interaction moved from the generic reciprocal prompt 'What about you?' to more personalised questions, including 'Do you have your best activities?' and 'When do you do your activities?' (TON-P11-S4; LCI-P11-S4/U045). In the displayed next turn, the chatbot listed its activities and returned a question; P11 answered, 'I also like playing football' (U045). This example illustrates change during P11's learning period through reported adaptation and more personalised questioning, with local influence evident in the chatbot's response. It does not establish sustained ownership or a general developmental trajectory. Because comparable reflective accounts were limited for other learners, *self-reflectiveness* in ownership claiming represents a focused pattern rather than a uniform finding across participants.

4.3. Challenges and barriers to agency

Challenges to agency arose from limitations in personal, contextual, and relational resources. Personal barriers concerned what learners could understand, formulate, regulate, and adapt in real time. Contextual barriers arose from the task, technology, and learning environment. Relational barriers concerned the distribution of authority between learner and chatbot. These lenses can overlap: unfamiliar chatbot language is a contextual demand, its comprehension involves personal resources, and its effect on questioning rights can be relational. Teacher assistance, subtitles and clarification also enabled action, as the following examples show.

4.3.1. Personal barriers

Personal barriers constrained agency by limiting what learners could understand, formulate, and express while managing the emotional demands of speaking. Linguistic

constraints were particularly visible when learners encountered unfamiliar vocabulary or could not understand complete chatbot questions. P04 identified 'not knowing how to say a word' as a recurring difficulty and responded during the interaction with 'I don't know', 'Use simple words', and 'Please say it again' (SRI-P04-S2; TON-P04-S2; LCI-P04-S2/U019). P08 similarly answered only the part of a question that he understood and used Chinese or 'I don't know' for the remainder (TON-P08-S2; LCI-P08-S2/U018). Difficulties also arose from cognitive processing rather than individual words alone. P06 could hear separate words but could not always integrate them into sentence meaning; she therefore referred to the preceding question, inferred possible connections, and requested repetition or definitions (SRI-P06-S2; TON-P06-S2; LCI-P06-S2/U020). P10 required extended time to organise and formulate relevant responses (TON-P10-S3; LCI-P10-S3/U028). These linguistic and processing constraints reduced learners' capacity to respond spontaneously and made their participation dependent on simplification, clarification, or additional planning time. Emotional disruptions were less extensively documented but sometimes intensified these difficulties. P01 became flustered after noticing a problem with her 'although … but … ' construction and continued only after teacher reassurance (TON-P01-S2; LCI-P01-S2/U014). P03 and nearby peers laughed after a recognition error, paused, and then repeated the answer, while P02 described becoming irritated and distracted when he could not continue (TON-P03-S2; LCI-P03-S2/U015; SRI-P02-S2). Personal barriers therefore narrowed the range of actions immediately available to learners, even when they subsequently exercised agency through repair or compensation.

4.3.2. Contextual barriers

Contextual barriers received the broadest support across the datasets and arose from task requirements, chatbot limitations, inaccessible output, and dependence on external scaffolding. The English-only and no-translation condition in oral interactions substantially increased the interactional burden. P04 relied on 'I don't know' and 'Use simple words' and repeatedly sought teacher confirmation, while P06 encountered unfamiliar vocabulary and requested repetition or definitions (TON-P04-S2; LCI-P04-S2/U019; TON-P06-S2; LCI-P06-S2/U020). P10 similarly reported that the chatbot used only English and could not provide the classroom-style Chinese explanation he needed; his interaction consequently depended on fallback responses and topic redirection (SRI-P10-S2; TON-P10-S2; LCI-P10-S2/U021). The fixed topic and duration also oriented participation towards completion. More than half of participants reported no personal goal beyond sustaining the required interaction for the assigned period (SRI-P02-S2; SRI-P04-S2; SRI-P05-S3; SRI-P07-S1; SRI-P10-S4; SRI-P11-S2). Technological constraints further disrupted learners' ability to maintain or redirect the conversation. P01 repeated a misrecognised question twice before abandoning the unsuccessful repair and asking a different task-related question (TON-P01-S3; LCI-P01-S3/U015). P09 similarly introduced a new question after the chatbot reported that it could not respond (TON-P09-S3; LCI-P09-S3/U032). These repairs maintained participation, but the chatbot failure had already restricted the direction the learner originally intended to pursue. Chatbot output created additional access problems through unfamiliar vocabulary and excessive speech rate. P09 reported relying on subtitles because the chatbot spoke quickly and used unfamiliar words (SRI-P09-S1), while P10 could not adjust its language through Chinese clarification (SRI-P10-

S1). Learners therefore depended on resources outside the chatbot interaction. P02 sought teacher translation before using the Chinese place name Zhongshan Ling, P03 requested vocabulary assistance from a peer, and P04 sought teacher confirmation while monitoring the remaining task time (TON-P02-S1; LCI-P02-S1/U013; TON-P03-S2; LCI-P03-S2/U015; SRI-P04-S2; TON-P04-S2). External scaffolding enabled learners to continue, but it also showed that their capacity to exercise agency independently remained conditional on task design, chatbot accessibility, and available classroom support.

4.3.3. Relational barriers

Relational barriers arose from the chatbot's recurrent control of questions, topics, and subsequent turns, which often positioned learners as respondents rather than conversational owners. P06 produced relevant and linguistically developed answers but asked no questions, leaving each topic transition to the chatbot (TON-P06-S3; LCI-P06-S3/U029). P10 showed the same pattern in Session 4 despite adopting vocabulary from earlier chatbot turns and tolerating recognition errors (TON-P10-S4; LCI-P10-S4/U042). These interactions demonstrate that competent participation did not necessarily provide control over conversational direction. The pressure created by recurrent questioning was made explicit when P04 complained that the chatbot was 'asking so many questions' and began recycling its questions back to it (TON-P04-S1; LCI-P04-S1/U018). P07 similarly used 'I don't know, what about you?' to transfer responsibility for the next contribution to the chatbot (TON-P07-S1; LCI-P07-S1/U012). Such moves attempted to renegotiate an unequal distribution of questioning rights. Other learners achieved greater reciprocity by rejecting unwanted topics or posing increasingly specific questions, as illustrated by P03 and P11 (TON-P03-S3; LCI-P03-S3/U026; TON-P11-S4; LCI-P11-S4/U045). The contrast indicates that relational asymmetry was negotiable but remained consequential: when learners did not challenge the chatbot-led sequence, fluent task performance could coexist with limited conversational ownership. Appendix A (Figure A1) summarises the codes developed for RQ1 and RQ2 through an abductive analysis that moved iteratively between the conceptual framework and the three linked datasets. It shows how theory-informed codes were refined against patterns in the learner-chatbot interactions, stimulated-recall interviews, and teacher-observation notes to produce the final agency configurations and resource-constraint categories.

5. Discussion

The findings show that young adolescent EFL learners exercised agency in two types distinguished by who determined the interaction's overarching intentionality. Within teacher-assigned goals, reactive agency took the form of task fulfilment as learners planned, regulated, adapted, and evaluated their participation to sustain the conversation. When learners introduced purposes of their own, proactive agency emerged through boundary testing and ownership claiming, enabling them to investigate the chatbot, resist unwanted directions, select topics and questions, and redistribute conversational control. Across these types, evidence for forethought and self-reflectiveness was uneven, whereas regulatory action was more extensively documented. Learners' agency was constrained by personal linguistic, processing, and emotional demands; contextual conditions involving the task, technology, and external support; and relational

asymmetries created by the chatbot's recurrent control of questions, topics, and turns. These constraints shaped whether learners could merely sustain participation or influence the conversation's direction.

5.1. How learners enacted reactive and proactive agency

Reactive agency was enacted through the purposeful regulation of task fulfilment. Learners prepared for the activity, anticipated difficulties, sought assistance, adjusted their language, managed recognition problems, requested clarification, and replaced unsustainable conversational directions. Although the teacher established the topic and duration, learners still controlled how they proceeded. Bandura's (2001, 2006) framework explains this regulation: *forethought* appeared in preparation and fallback strategies, *self-reactiveness* guided adjustments, and *self-reflectiveness* informed evaluations of problems and responses. The finding therefore locates agency within teacher-structured activity, where it might otherwise be interpreted simply as engagement or compliance. Although learners did not select the entire task, they exercised control over how they pursued it.

Repair provides the strongest connection between reactive agency and existing process research. Choi (2026) showed how L2 learners used third-position repair when chatbot responses exposed misunderstanding, while Choi and Oh (2026) traced changes in learners' turn-taking practices across repeated ChatGPT role plays. The present findings support the importance of repetition, clarification, reformulation, accommodation, and topic adjustment, but show that these practices serve different purposes. Some repairs recover an intended meaning or direction; others accommodate the chatbot so participation can continue. Repair is therefore not inherently agentic. Its significance depends on the learner's *intentionality*, the condition being addressed, and the conversational possibility created by the action. By linking interaction records with observation and recall evidence, the study contributes a purpose-action-consequence account of agency. Observable action establishes what happened, while the linked evidence helps explain why it was selected and what it enabled. Conversational smoothness is consequently insufficient evidence of learner control. A fluent exchange may remain chatbot-led, whereas a hesitation, refusal, or repair may alter its direction.

This distinction qualifies outcome-focused GenAI-speaking research. Previous studies have reported speaking gains, increased opportunities for practice, positive perceptions, willingness to communicate, and engagement (Guan, Zhang, and Gu 2025; Tai and Chen 2024, 2025; Wang, Li, and Zou 2025; Wu, Hapsari, and Huang 2026). These findings establish the potential value of designed chatbot activities, but provide less evidence about who organises the conversation. The present study shows that learners may participate successfully while responding mainly to a chatbot-led agenda. It therefore adds conversational control as a consideration alongside performance, participation, and affective response.

Proactive agency emerged when learners introduced a purpose that exceeded task maintenance. Boundary testing positioned the chatbot as an object of inquiry whose flexibility, knowledge, restrictions, and responses could be examined. This finding extends Liu, Darvin, and Ma's (2025) account of usefulness developing through experimentation and Lai et al.'s (2026) conception of agency as differentiated engagement with GenAI. The present study shows how younger learners enact such negotiation through spoken action. Its contribution must nevertheless remain cautious: boundary testing

demonstrated curiosity and temporary control, but it did not always develop into sustained critical evaluation or language learning.

Ownership claiming enacted proactive agency through attempts to shape conversational direction. Learners rejected unwanted directions, introduced alternatives, posed reciprocal or follow-up questions, and redistributed responsibility for subsequent turns. This finding adds a relational dimension to Jeon's (2024) account of young learners' differing perceptions of chatbot affordances. Control has a sequential character because a learner-generated direction becomes consequential only when the chatbot recognises and responds to it. Agency therefore concerns the distribution of rights to question, select topics and resist directions. Lai et al. (2026) distinguish performance proxy, learning proxy, spatial, collective, and shared agency according to how learners use and coordinate GenAI. The present study contributes to this differentiated account by addressing a more specific interactional question: whose purpose organises the spoken interaction? Its answer distinguishes task fulfilment under a teacher-defined purpose from learner-generated boundary testing and ownership claiming.

5.2. How personal, contextual, and relational resources enabled and constrained agency

The second research question examined how resources and their limitations shaped learners' agency. Personal constraints narrowed what learners could understand and formulate, contextual constraints structured the support and options available, and relational constraints affected who controlled questions, topics, and subsequent turns. These constraints shaped the form of agency learners could exercise and the degree of conversational control they could retain.

Personal constraints affected learners' capacity to act in real time. Limited vocabulary, processing demands, uncertainty about grammatical form, and insufficient formulation time restricted spontaneous responses and made it harder to retain an intended direction. Learners responded by simplifying, clarifying, redirecting, accommodating the chatbot, or seeking assistance. This finding supports Jeon's (2024) argument that chatbot affordances depend on learners' abilities and perceptions. Extended chatbot output may provide useful practice for one learner while imposing an excessive processing demand on another. The present study extends this account by showing how personal resources affect the actions available after a difficulty occurs.

Contextual resources shaped whether these regulatory actions were possible and effective. Assigned topics and duration requirements provided a structure for preparation and participation, while teacher guidance, peer assistance, subtitles, repetition, and reassurance could restore an interrupted conversation. This pattern is consistent with research showing that young learners' GenAI-speaking experiences depend on task design, language support, and pedagogical guidance (Tai and Chen 2024, 2025). Min, Gao, and Zhang (2026) similarly found that teacher guidance and task mode influenced behavioural, cognitive, and social engagement. However, the present findings reveal a tension within such support. Structure can make a speaking activity manageable, yet it may orient learners towards completing an assigned requirement rather than developing conversational purposes of their own. Technological conditions created further constraints when recognition errors, inaccessible output, topic drift, or system refusals

restricted an intended direction. Zou et al. (2026) likewise found that high interactivity was not uniformly advantageous when feedback overload, recognition problems, and conversational disruption weakened engagement. The current study adds learner control to this design problem: conversational openness provides limited practical choice when learners cannot understand the chatbot, recover an intended topic, or formulate an alternative direction. Under these conditions, openness increases the linguistic and regulatory work required to sustain learner-generated action.

Relational constraints arose from the distribution of conversational initiative between learner and chatbot. The chatbot's recurrent role as questioner and topic initiator could sustain the exchange while positioning learners primarily as respondents. Negotiating this asymmetry required sufficient linguistic resources and a learner-generated purpose, but an attempted redirection became consequential only when the chatbot recognised and followed it. This finding develops the situated accounts offered by Lai et al. (2026) and Schoots-Snijder, Tigelaar, and Admiraal (2026). Lai et al. showed that agency with GenAI depends on the coordination of personal, human, and technological resources, while Schoots-Snijder et al. demonstrated that secondary students mobilise agentic features and environmental resources in uneven ways. The present study identifies a further relational consequence: resources affect whether learners can ask the next question, retain a preferred topic, resist a transition, or redistribute responsibility for the conversation. The chatbot's response therefore conditions the effect of learner action, although the learner remains the source of the purpose being pursued.

These constraints shaped whether learners could maintain task participation, investigate the chatbot, or influence conversational direction. Access provided an opportunity to speak; conversational control required sufficient resources to make a learner-generated purpose consequential. Compliance, proficiency, novelty, teacher presence and the app's recurrent questioning offer alternative explanations for the observed actions; the present design does not isolate their effects. Purposeful selection and conversational consequence therefore support an agency interpretation only at the level evidenced by each dialogue round.

5.3. Implications

The answers to both research questions add learner control as a criterion for evaluating GenAI-speaking activities. Min, Gao, and Zhang (2026) found that teacher guidance and task mode shaped engagement, while Zou et al. (2026) showed that high interactivity was not uniformly superior. The present findings suggest testing designs that combine linguistic and procedural support with genuine opportunities for learners to influence what happens next. For teachers, the reported clarification, repair and reciprocal-questioning dialogue rounds suggest rehearsing these moves and allowing learners to propose topic alternatives. For app designers, the reported language-access and topic-retention difficulties suggest testing manageable turn length, accessible clarification and reliable topic return.

5.4. Limitations and future research

Several limitations bound the answers to both research questions. The study involved a small group from one school, one application, and a teacher-structured routine; teacher recommendation and phone ownership restrict transferability. Screen and

audio records were checked manually, while displayed ASR errors were retained as part of the live interaction; the comparison could not reconstruct how the chatbot interpreted a learner's intended wording. Interpretations of intentions, anticipations, and evaluations relied on recall accounts that may have been affected by memory, interview prompts, and the first author's dual role; observer presence, leading prompts and translation decisions may also shape the accounts. Changes across interactions may also reflect familiarity, topic differences, support, or reduced novelty. The evidence does not establish progression from reactive to proactive agency, effects on speaking proficiency, or stable learner characteristics. The aggregate thematic presentation does not provide systematic within-case and cross-case trajectories, and findings may depend on the app version and settings in use.

Future research should vary topic choice, preparation, teacher support, chatbot turn length, clarification functions, and the balance between chatbot and learner questions. Longer studies linking verified audio, interaction records, observation, and indexed recall could examine whether control strategies transfer across topics and platforms and when boundary testing or ownership claiming becomes sustained.

6. Conclusion

In this small, selected sample, purposeful task regulation and attempts to redirect the chatbot show why speaking participation and conversational control should be examined separately. Reactive and proactive agency describe situated orientations within the reported dialogue rounds, not fixed learner types or a demonstrated progression. Their value lies in connecting learner purpose with action and chatbot uptake while recognising the resources and evidence needed to establish that connection.

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Author contributions

CRediT: **April Jiawei Zhang:** Data curation, Formal analysis, Investigation, Methodology, Resources, Visualization, Writing – original draft, Writing – review & editing; **Wei Wei:** Conceptualization, Data curation, Formal analysis, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Declaration of generative AI use

During manuscript preparation and revision, the authors used OpenAI ChatGPT to assist with language editing, structural revision, and reference-format checking. The authors reviewed and verified all generated material and take full responsibility for the content of the manuscript.

Data availability statement

The anonymised learner-chatbot interaction transcripts supporting the findings of this study are available from the corresponding author upon reasonable request and subject to applicable institutional ethics and safeguarding requirements. Raw audio, screen recordings, and identifiable application records are not publicly available because they contain identifiable data from minor participants.

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Appendix A

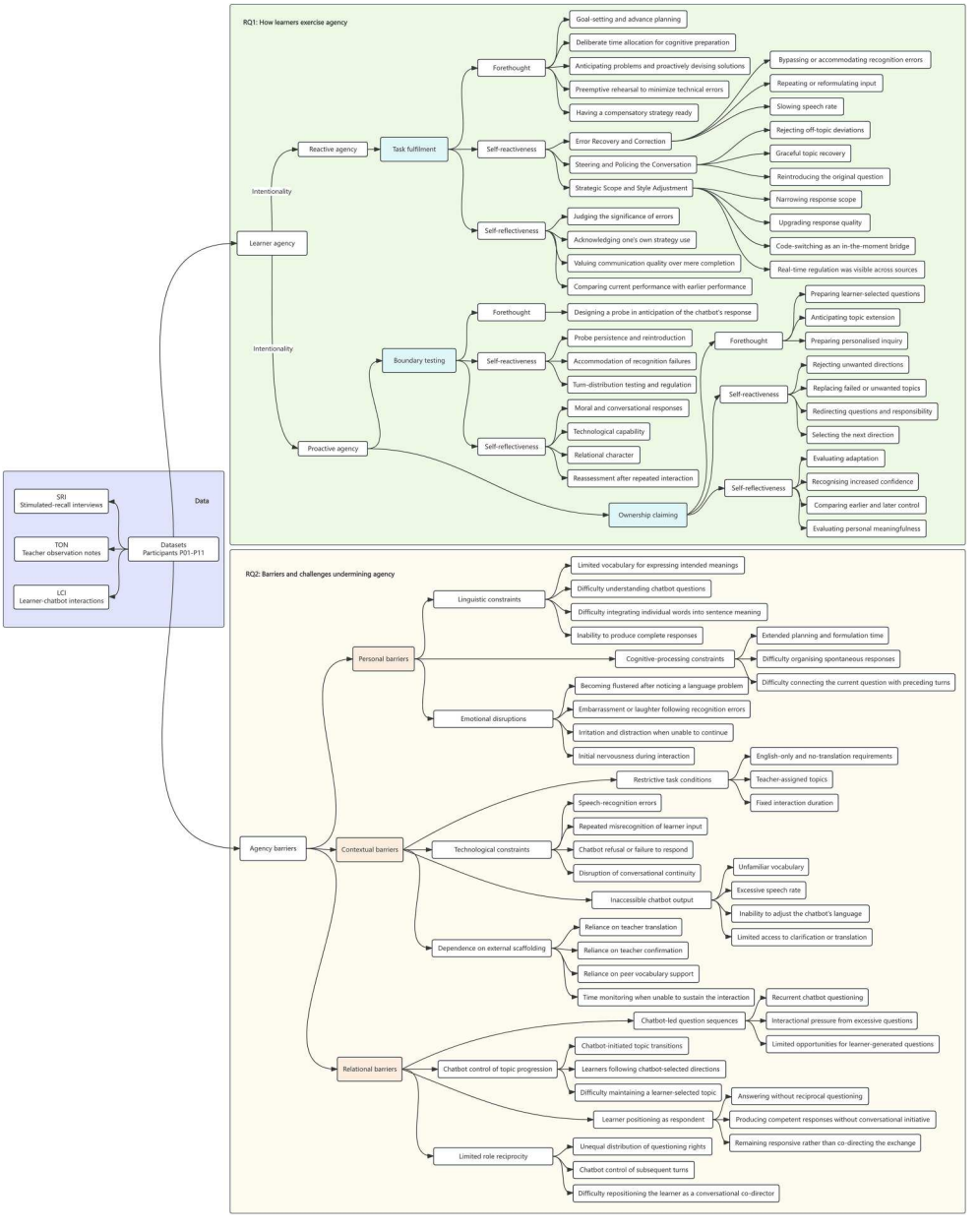


Figure A1. Coding framework for agency configurations and resource constraints Coding framework for agency configurations and resource constraints.